Informatique et systèmes de communication ISC

Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

Signal and information (SigInf)

204.1

Dr. Robin Delabays and Dre. Benedetta Franceschiello
benedetta.franceschiello@hevs.ch
Robin.delabays.hevs.ch

0. PREVIOUSLY ON SIGINF

What is a signal

Signal = A signal is essentially a physical quantity that varies over time, space, or another independent variable, and it conveys some form of information. Signals can represent audio waves, images, light intensity, radar echoes, physiological data (like heartbeats), or even stock prices.

Définition

Recap part 1

Signal Types:

- **Continuous** vs. **Discrete** signals: Continuous signals are defined over all time, while discrete signals are defined at intervals.
- Important signals: **Impulse**, **Unit Step**, and **Sinusoids**.

Signal Characteristics:

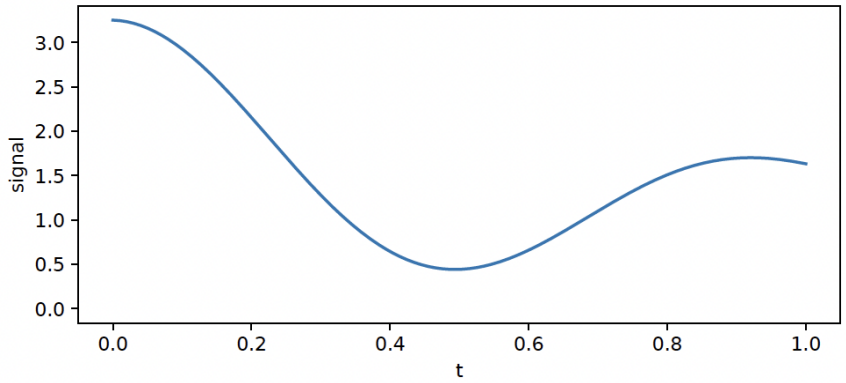
- **Energy** and **Power**: Measure the strength of a signal; energy is used for finite signals, and power for continuous ones.

Noise:

- **Definition**: Unwanted random variations in the signal.
- **Signal-to-Noise Ratio (SNR)**: A metric to assess signal clarity.

Sampling:

- **Nyquist-Shannon Theorem**: The sampling rate must be at least twice the signal's highest frequency to prevent aliasing.



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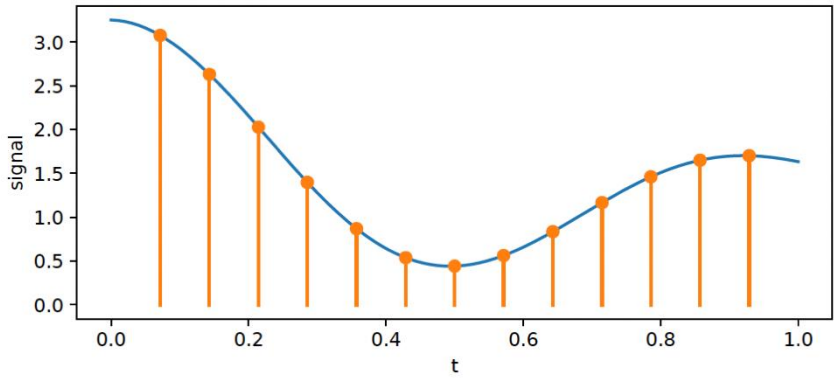

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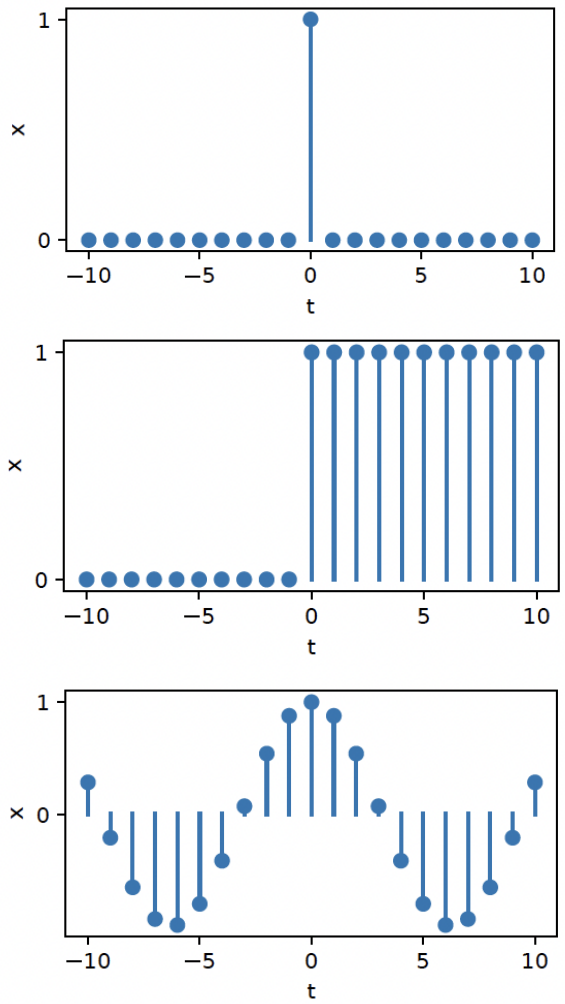

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$$E(x) = \sum_{k=1}^{T} (x[k])^2 \delta_t \, .$$

$$\mathrm{SNR} = \frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}} = \frac{\sum_k (x[k])^2}{\sum_k (\xi[k])^2} \, .$$

Recap part 2

Filtering: Removes unwanted frequency components (e.g., noise).

- **Low-Pass**: Passes low frequencies, blocks high.
- **High-Pass**: Passes high frequencies, blocks low.
- **Band-Pass**: Isolates a range of frequencies.

Windowing: Isolates parts of a signal in the time domain to reduce frequency leakage: **Rectangular**, **Hann**, and **Hamming**.

Smoothing (Sliding Average): A simple method to reduce short-term fluctuations in data.

Fourier Transform: Converts time-domain signals to the frequency domain, essential for analyzing frequency content and designing filters.

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Lesson 7. Statistical Signal Processing

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1. FIRST-LEVEL ANALYSIS

Purpose of Statistical Signal Processing

Why?

1. Noise reduction, to **enhance signal quality** and extract meaningful information
2. **Identify patterns** and trends within signals.
3. **Characterize** the signal property
4. If I'm able to interpret the signal, I can build **predictive models**

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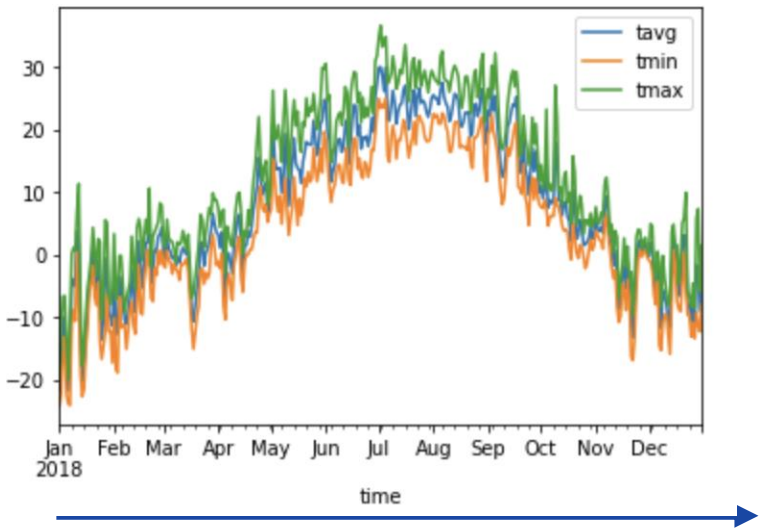
- Clean and filter
- I can study repetition
- I can **describe** the signal **(descriptive statistics)**
- Out of many observations, I can build predictions



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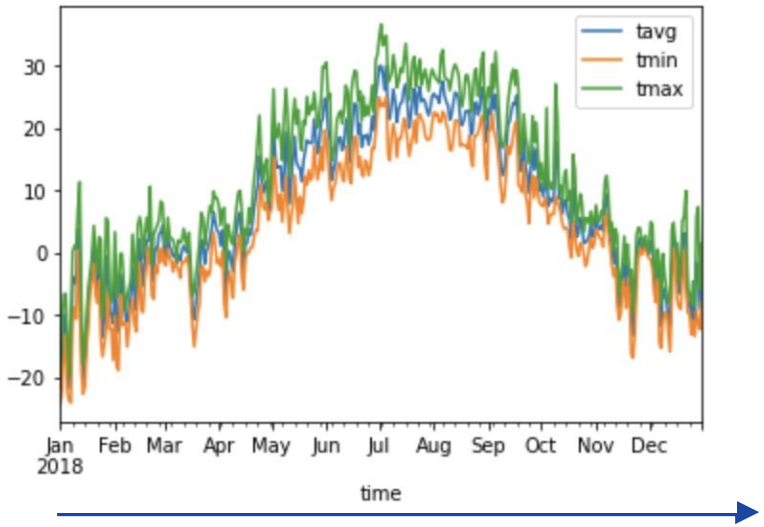



```
# You should install meteostat: pip install meteostat
# Link to check out: https://dev.meteostat.net/python/#example & https://dev.meteostat.net/bulk/#quick-start
# https://github.com/meteostat & https://dev.meteostat.net/python/api/point/#parameters
# Import Meteostat library and dependencies
```

Descriptive statistics of a time series

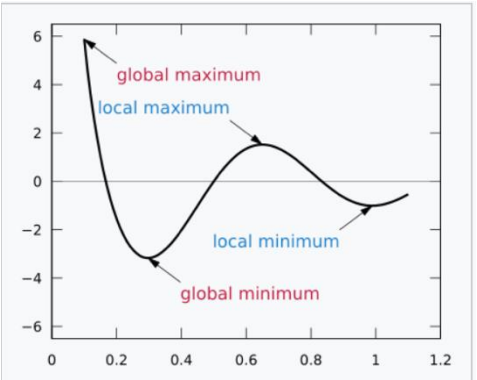
Signal characterization:

1. **Max, Min, Mean** - Central value of the signal.
2. **Median**: Middle value when sorted.
3. **Variance & Standard Deviation**: Measure of signal spread (and fluctuation).
4. **Distribution** analysis: Shows patterns like normal or skewed distribution.

Descriptive statistics of a time series

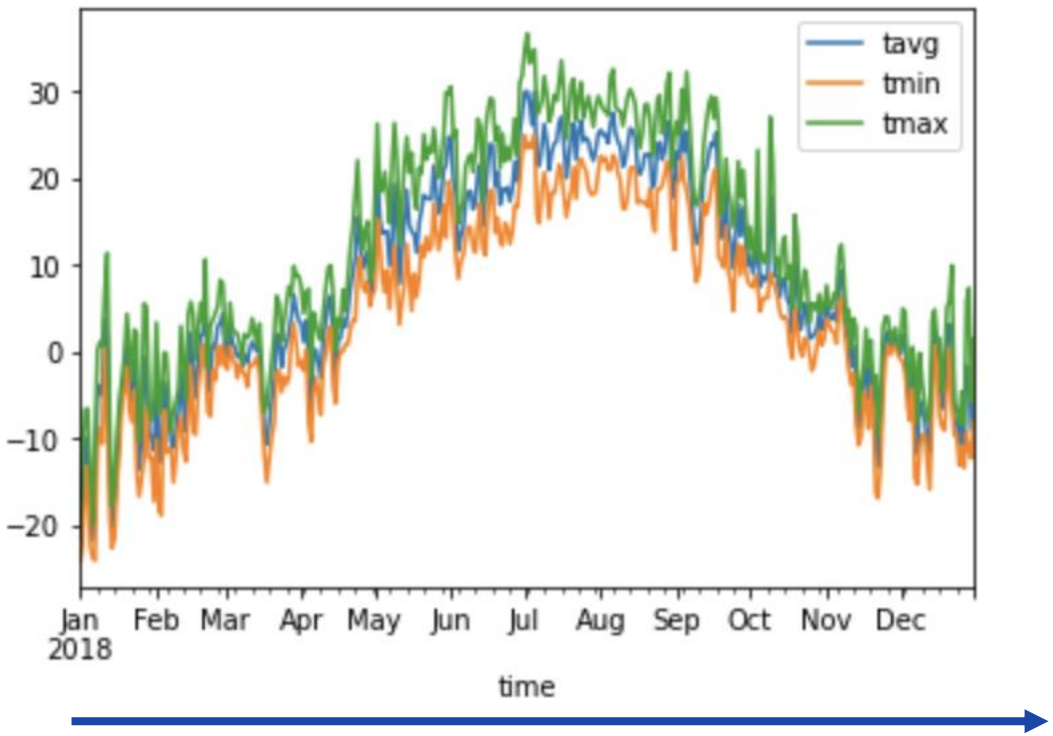
Baie Saint-Paul evolution of temperature over time:

- Sampling: 1 temperature point / hour
  - Plotting **Max** & **Min** temperature per day, per week or per month



  - Plotting **Average** temperature per day, per week or per month

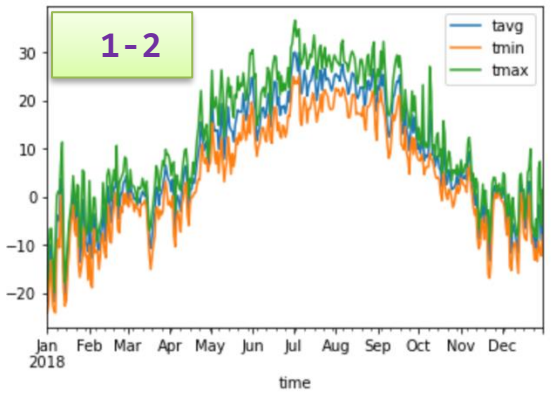
$$\bar{x} = \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)$$



Example

Baie Saint-Paul evolution of temperature over time:

1. **[Question 0] Do I need preprocessing? Filtering? De-noising?**
2. Plot and visualize to observe trends, seasonality and anomalies
3. Calculate summary statistics (mean, median, standard deviation).
4. **[Later] Calculate entropy, to assess predictability (Class 8-9)**
5. Look for patterns (aka repetitions): seasonal patterns, year patterns, etc.



4 $$H(\tilde{X}) = -\sum_{x_i} p(x_i) \log_2 p(x_i) = -\sum p(x) \log p(x)$$

How do I assess if my entropy is high or low? I compare it with the entropy from other years, to assess if the signal in 2018 is variable, or not. **Variability affects predictability.**

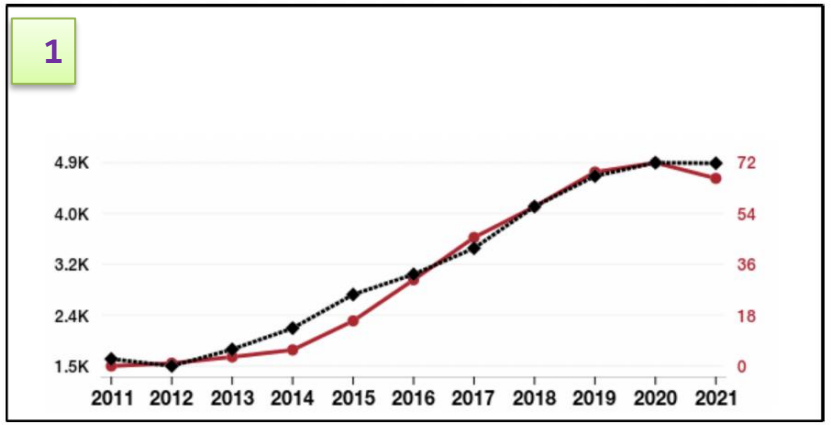
Correlation, stationarity and autocorrelation

Understanding time series structure:

1. **Correlation:** given two signals that have the "same shape", can we quantify to what extent they "look alike?"

2. Stationarity

3. Autocorrelation functions (ACF and Partial ACF)



Pearson's correlation coefficient:

$$r_{xy} = \frac{\sum_k (x[k]-\bar{x})(y[k]-\bar{y})}{\sqrt{\sum_k (x[k]-\bar{x})^2}\sqrt{\sum_k (y[k]-\bar{y})^2}},$$

$$\bar{x} = \frac{1}{T}\sum_{k=1}^{T} x[k], \quad \bar{y} = \frac{1}{T}\sum_{k=1}^{T} y[k].$$

Correlation, stationarity and autocorrelation

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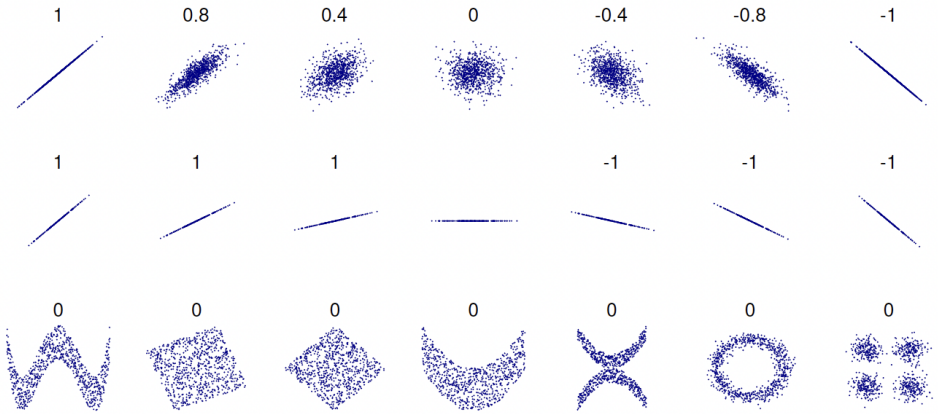
1 **Warning!** Correlation $\neq$ Causality $\neq$ Coincidence



Source: Wikipedia (23.4.2024)

Correlation, stationarity and autocorrelation

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3. Autocorrelation functions (ACF and Partial ACF)

2 If I consider again the meteo data, why would I make a signal highly non stationary, **stationary**?

Because in so doing I can isolate patterns:

- **Differencing**: Subtracting the previous day's temperature from the current day can help remove trends and make the data more stable over time.

- **Seasonal Differencing**: Subtracting the value from the same day in the previous year (e.g., temperature on July 1, 2021, minus temperature on July 1, 2020) can help remove yearly seasonal effects, making the series more stationary.

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3. **Autocorrelation functions** (ACF and Partial ACF): they measure the relationship between a time series and its lagged values, indicating how current values relate to past values

3 Temperature data are autocorrelated, because today's temperature is related to recent day. High autocorrelation at certain lags indicates patterns or dependencies over time, which is essential in understanding seasonal or recurring patterns.

• **Applications in Signal Processing**:**Periodicity**: Identifying repeating cycles in the data, like annual temperature cycles.

• **Lagged Relationships**: Understanding how past values influence current values helps in forecasting and detecting seasonal patterns.

$$R_{ff}(l) = \sum_{n=0}^{N} f(n)f(n-l).$$

Correlation, stationarity and autocorrelation

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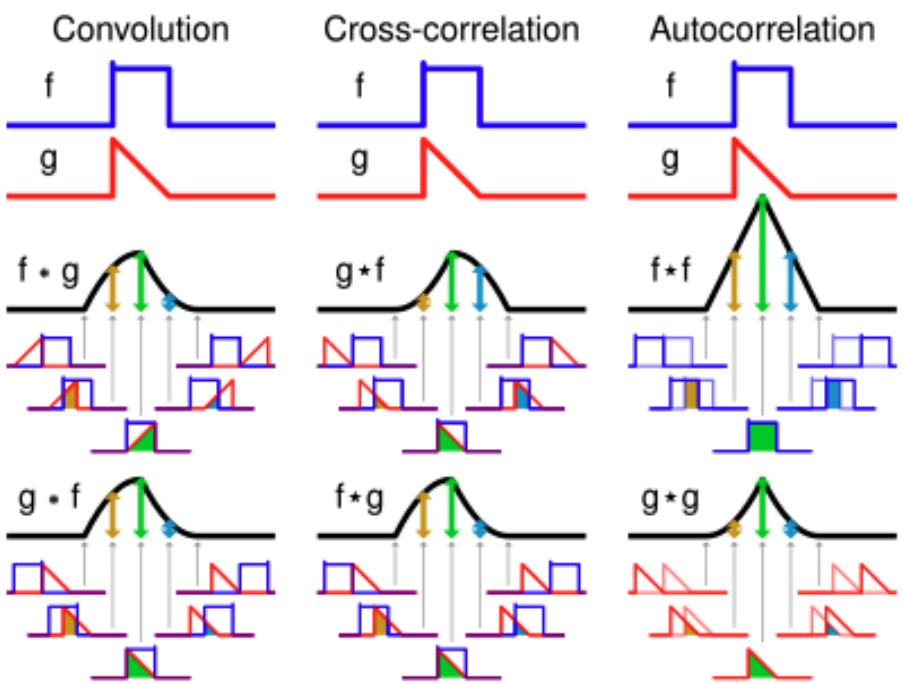
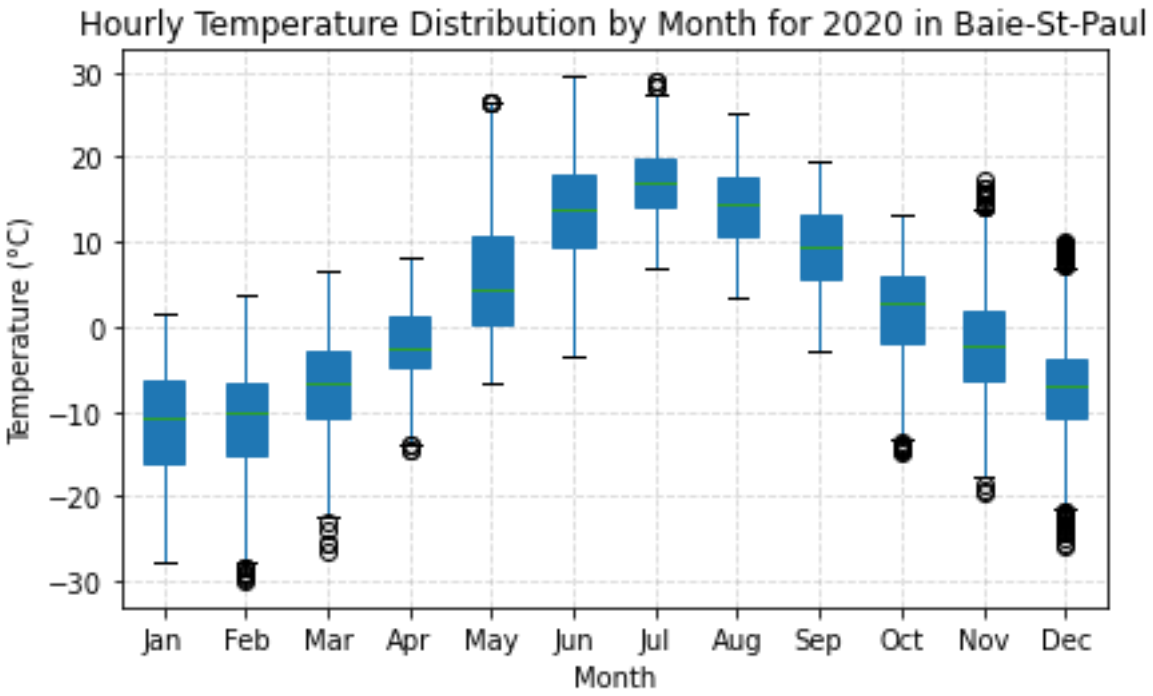
Disclaimer!!!



https://en.wikipedia.org/wiki/Autocorrelation

Box Plots are your friends

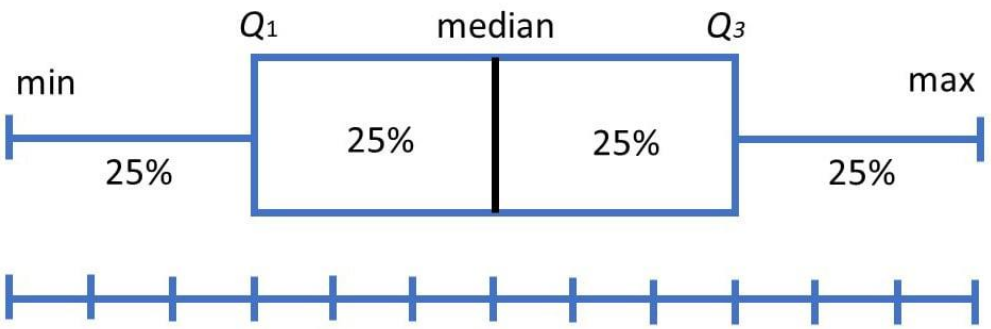


Analysis of Monthly Temperature

Box plots show the distribution of temperature each month:

- **Seasonal trends**: Observe typical temperature patterns by month (e.g. warmer in summer, cooler in winter).
- **Variability**: Wider boxes indicate higher temperature fluctuations.
- **Outliers**: Extreme values, like heatwaves or cold snaps, appear as outliers, they show **anomalies** in the time series

Box Plots: definition and interpretation



https://www.simplypsychology.org/boxplots.html

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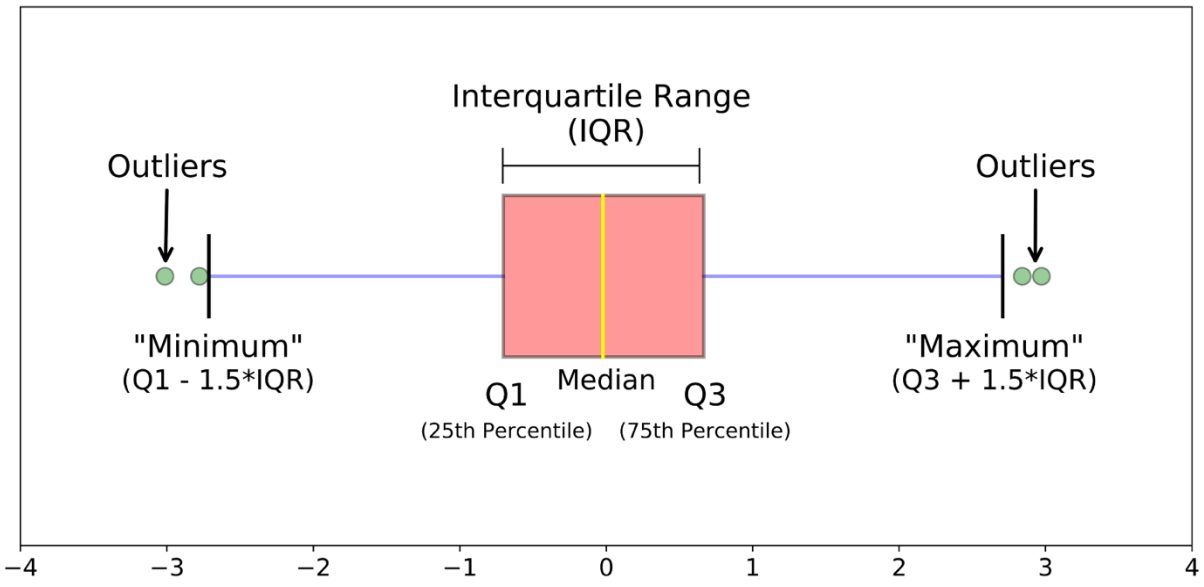

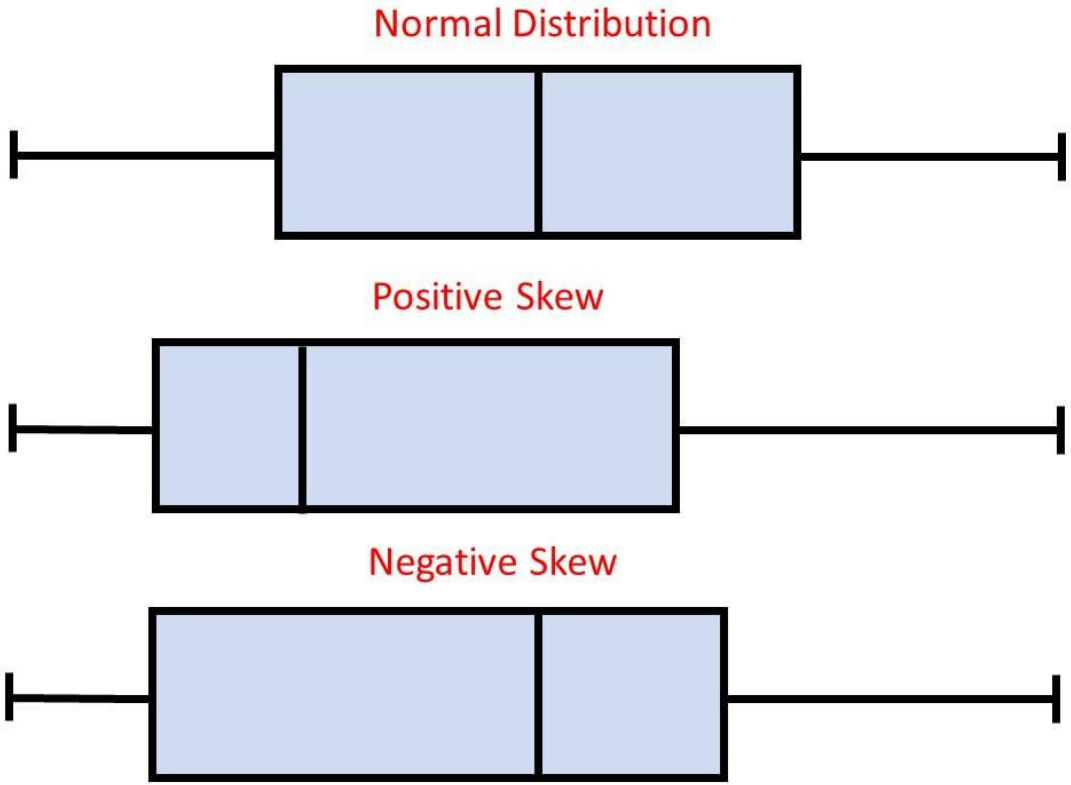

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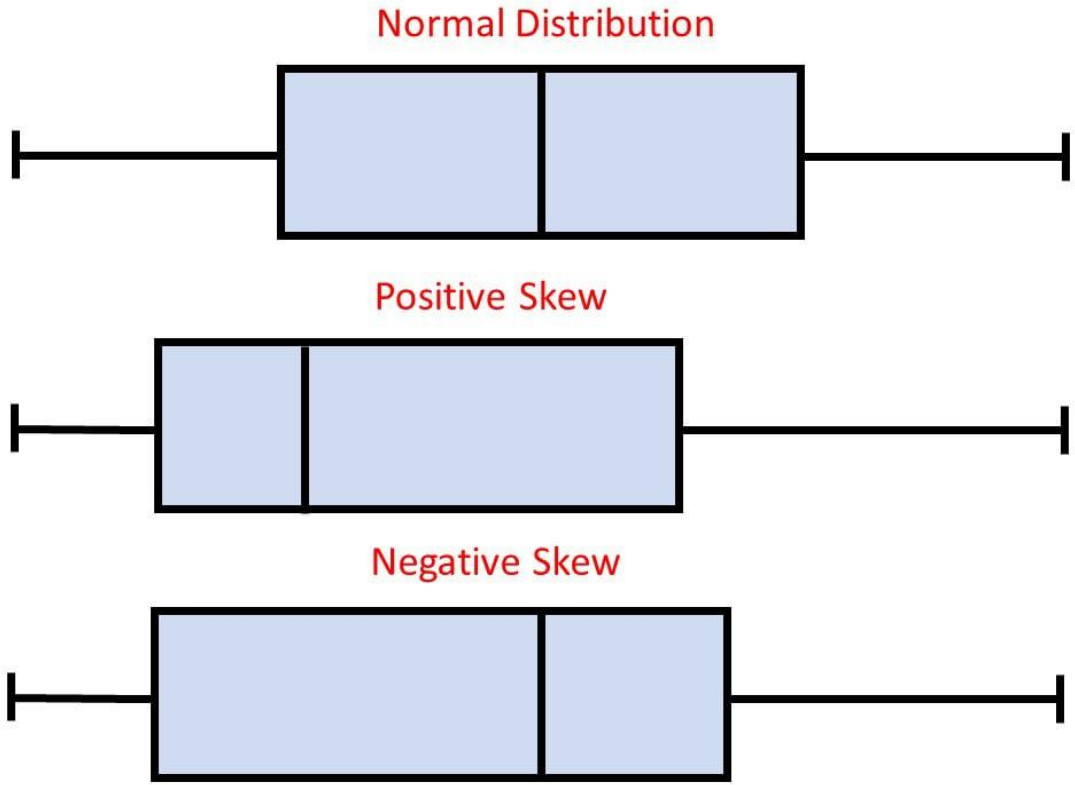
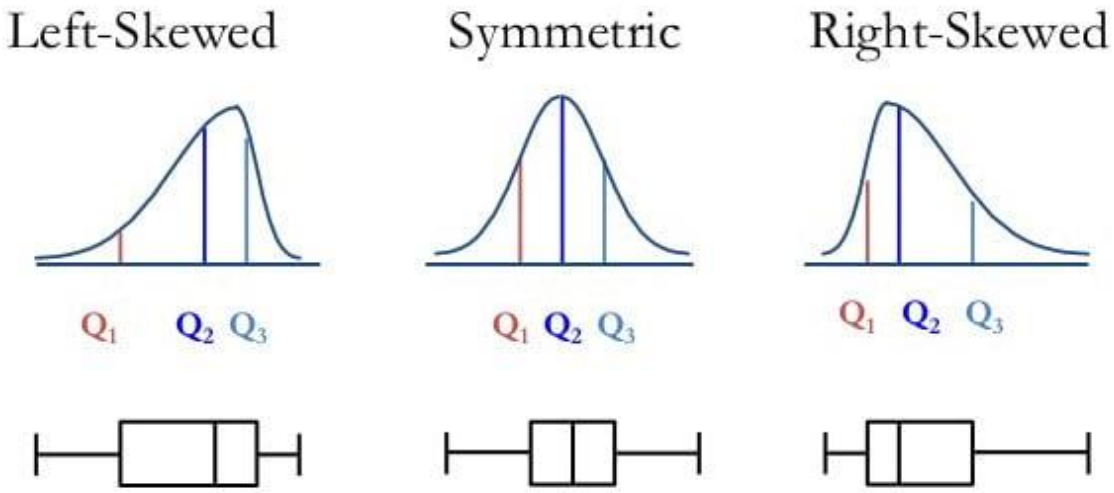
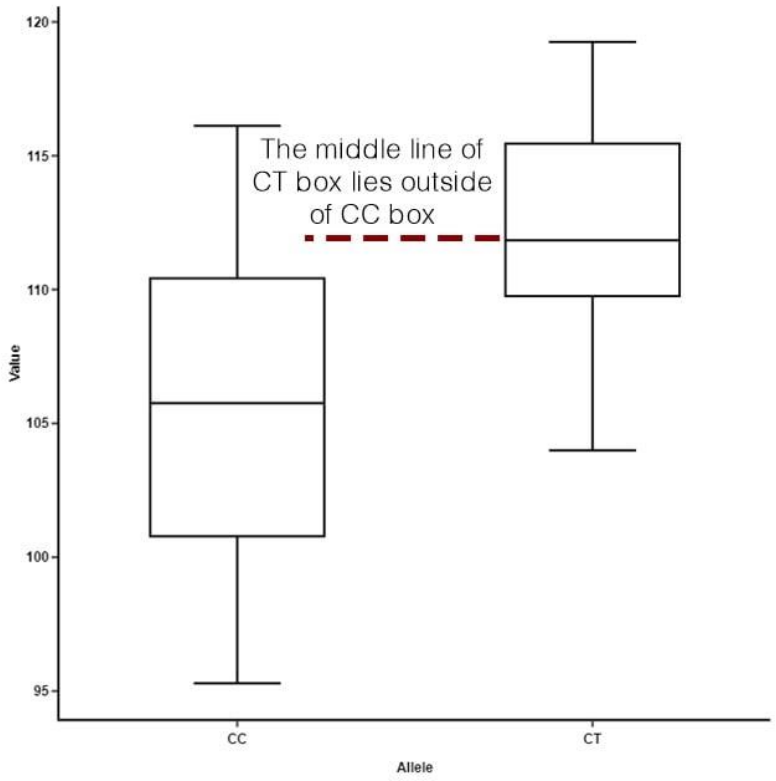
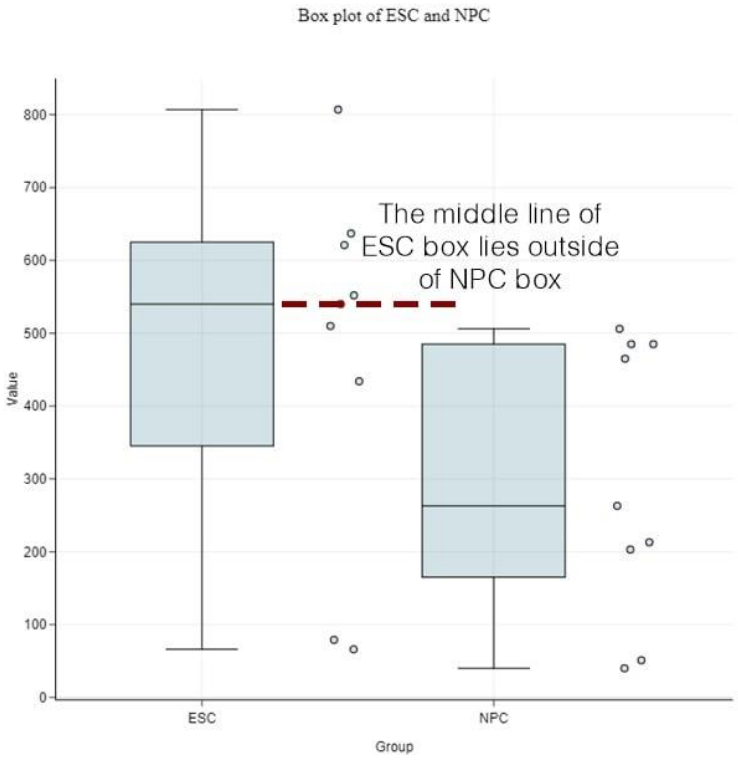




Example:

Compare the respective medians of each box plot. If the median line of a box plot lies outside of the box of a comparison box plot, then there is likely to be a difference between the two groups. This is when you run **statistics** to analyse the differences between distributions. (Parametric tests (t-test, d-Cohen), non-parametric tests (Wilcoxon Signed Rank Test)





Other ways to compute anomalies:

How the Z-Score Works

1. Definition of Z-Score:

1. The Z-score of a data point is calculated as:

$$z = \frac{x - \mu}{\sigma}$$

where:

1. x is the value of the data point.
2. μ is the mean of the time series.
3. σ is the standard deviation of the time series.

2. Interpretation:

1. The Z-score tells us how many standard deviations a particular value x is from the mean μ.
2. A **high absolute Z-score** (e.g., |Z| > 2 or 3) indicates that the value is far from the mean and is therefore likely to be an outlier (anomaly).

```
Outliers in May 2020 based on z-scores:
                     temp   z_score
time
2020-05-21 20:00:00  21.8  2.068384
2020-05-26 15:00:00  21.6  2.041948
2020-05-26 17:00:00  21.3  2.002293
2020-05-26 18:00:00  22.2  2.121257
2020-05-26 19:00:00  23.5  2.293094
2020-05-26 20:00:00  22.9  2.213785
2020-05-27 15:00:00  22.3  2.134475
2020-05-27 16:00:00  24.2  2.385621
2020-05-27 17:00:00  26.1  2.636767
2020-05-27 18:00:00  26.3  2.663204
2020-05-27 19:00:00  26.6  2.702859
2020-05-27 20:00:00  24.7  2.451712
2020-05-27 21:00:00  23.3  2.266658
2020-05-27 22:00:00  23.8  2.332749
2020-05-28 13:00:00  22.6  2.174130
2020-05-28 14:00:00  24.8  2.464931
2020-05-28 15:00:00  26.5  2.689640
2020-05-28 16:00:00  26.4  2.676422
2020-05-28 17:00:00  26.3  2.663204
2020-05-28 18:00:00  24.9  2.478149
2020-05-28 19:00:00  26.6  2.702859
2020-05-28 20:00:00  24.5  2.425276
2020-05-28 21:00:00  24.5  2.425276
2020-05-28 22:00:00  23.2  2.253439
2020-05-28 23:00:00  21.6  2.041948
2020-05-29 16:00:00  21.5  2.028730
2020-05-29 17:00:00  21.9  2.081603
2020-05-29 18:00:00  23.2  2.253439
2020-05-29 19:00:00  23.4  2.279876
2020-05-29 20:00:00  23.4  2.279876
2020-05-29 21:00:00  24.6  2.438494
2020-05-29 22:00:00  24.4  2.412058
2020-05-29 23:00:00  22.7  2.187348
```

Forecasting: who doesn't want to predict the future?

Naïve forecasting:

1. **Observation** of historical data patterns
2. **Assumption:** it assumes that the last observed value are the best predictor of the next value (e.g. if today it's 25°, tomorrow the temperature it's likely to be the same)
3. **Suitable for:** autocorrelated time series, with no fluctuations
4. Can you think about the limitations?

Moving average:

1. **Observation** of historical data patterns
2. **Methods:** calculates the average of the last "n" observations to smooth out fluctuations (e.g. I use the average temperature of the past 7 days to predict tomorrow's day temperature).
3. **Suitable for:** reducing noise in my date due to fluctuations.
4. Can you think about the limitations?

Forecasting: who doesn't want to predict the future?

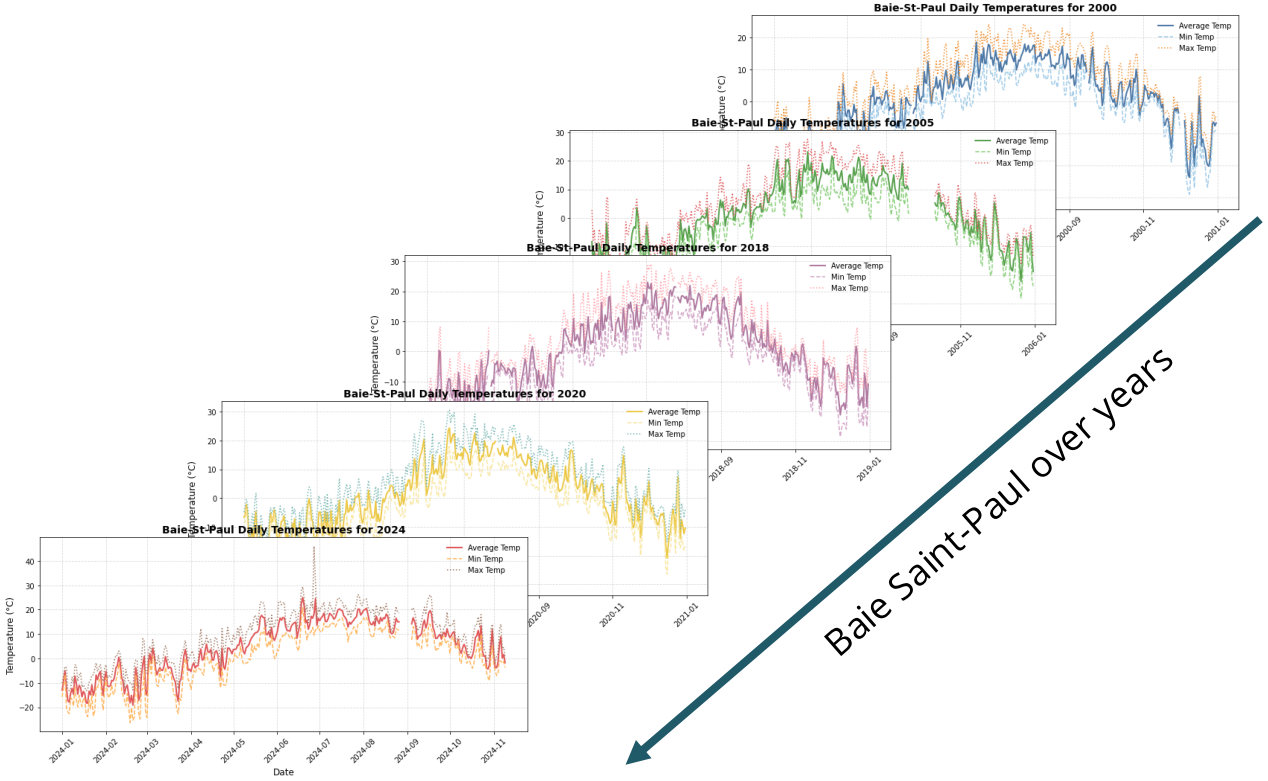
Take home message = Choosing the right forecasting method depends on the **data's characteristics**, such as trend, seasonality, and autocorrelation, etc. Always look at your data and be critical.

Remind

1. SECOND-LEVEL ANALYSIS

Second-Level Analysis: Group Analysis of Multiple Signals

Analyzing patterns across a group of signals (multi-dimensional)



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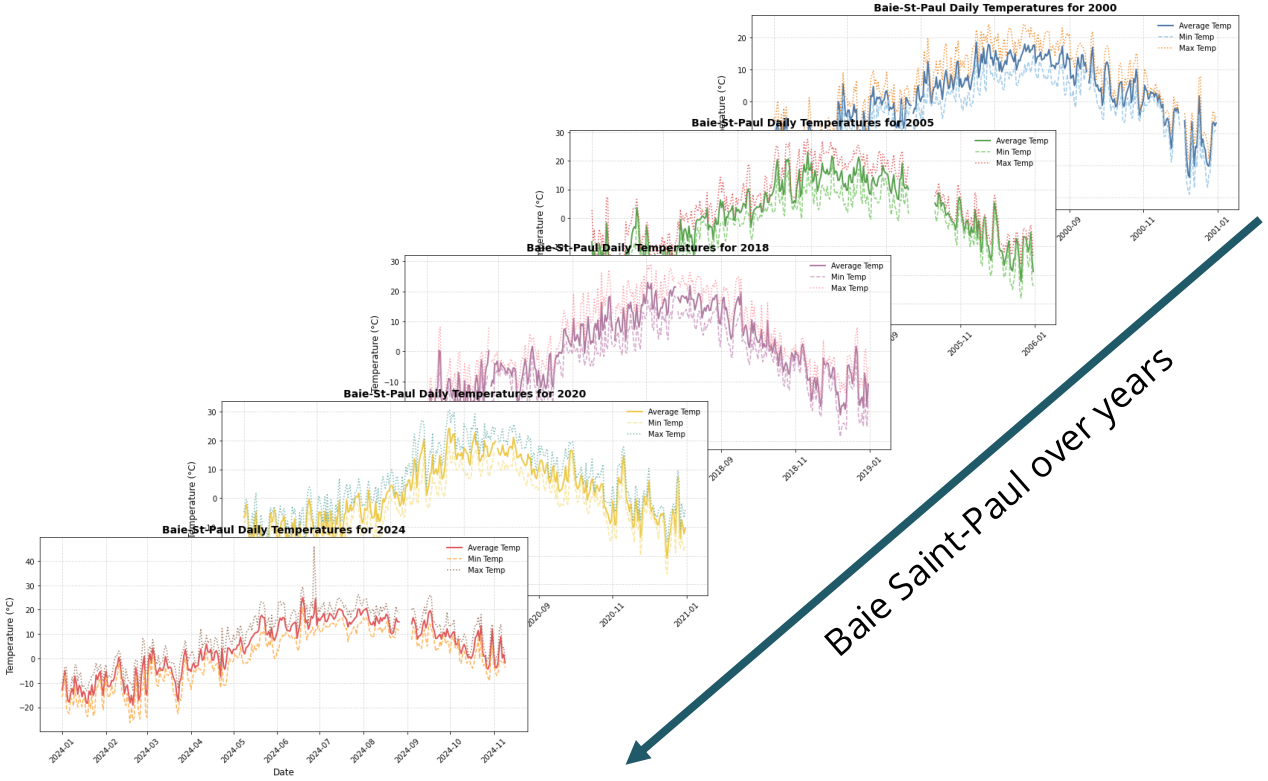

Baie Saint-Paul over years

All Quebec's cities over years

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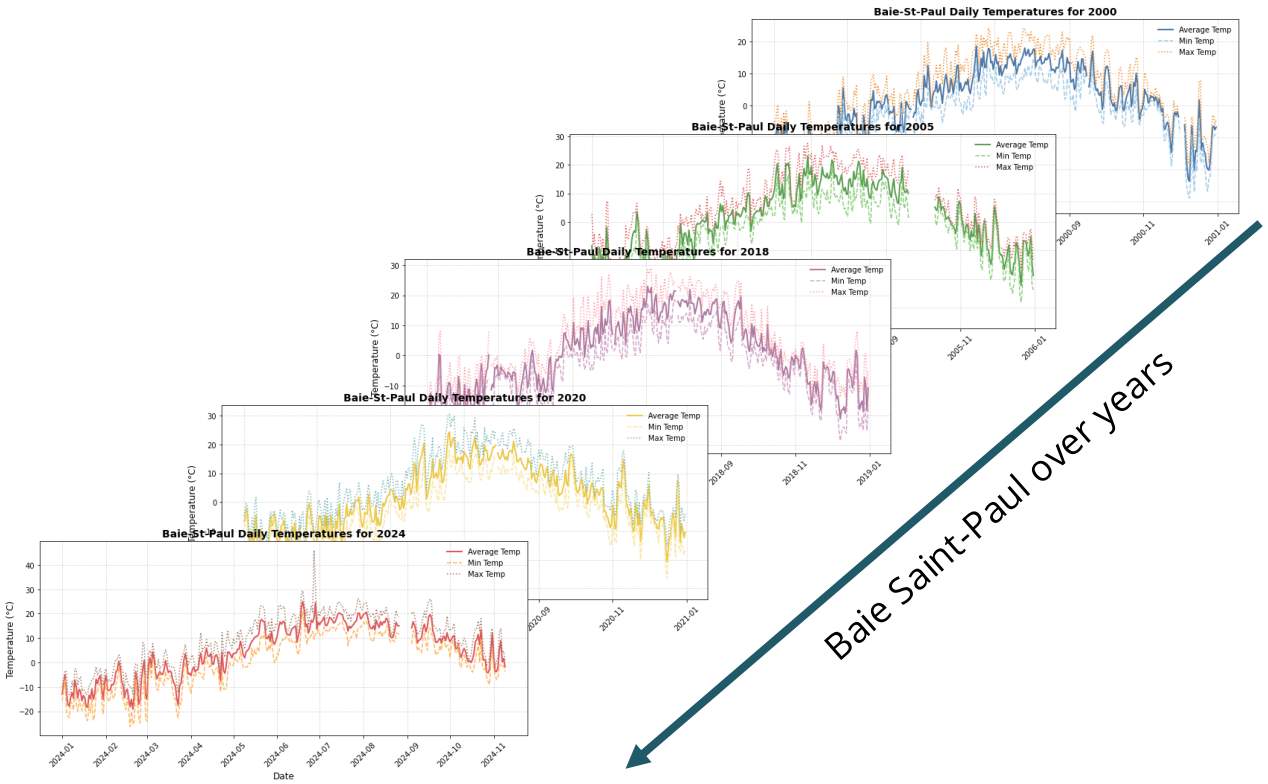
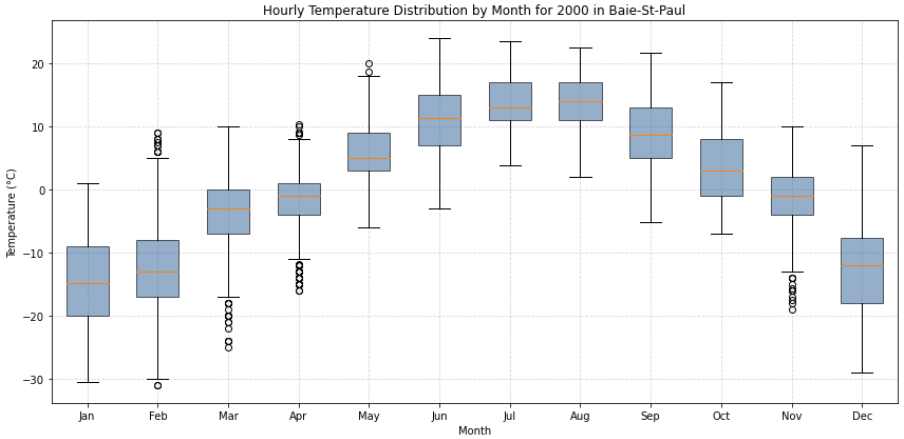
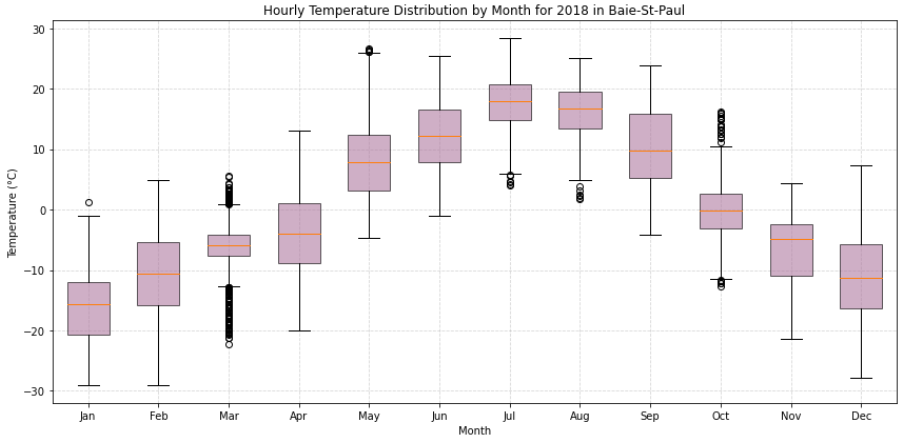
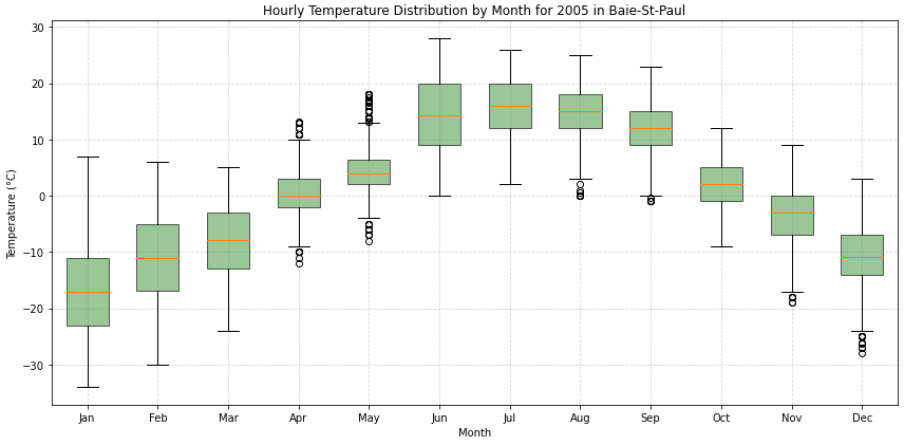
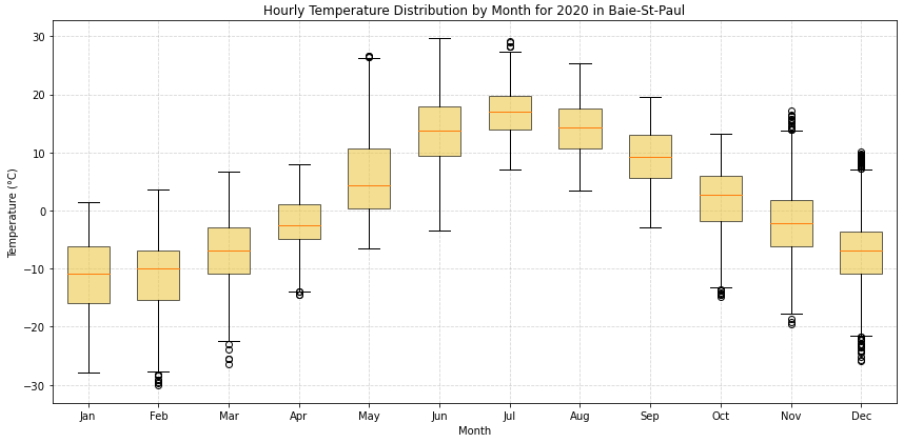


Baie Saint-Paul over years

All Quebec's cities over years

Descriptive statistics still applies:

- **Aggregated analysis** (e.g., average monthly values across cities).
- **Comparative trends for individual** (across time) vs. **group** (regional, national) behaviors.
- **Clustering** to identify similar behaviors within sub-groups / across groups.

Second-level analysis: box plots

Year-over-Year Comparison: Observe shifts or changes in distribution over years (detect potential trends or climate changes)